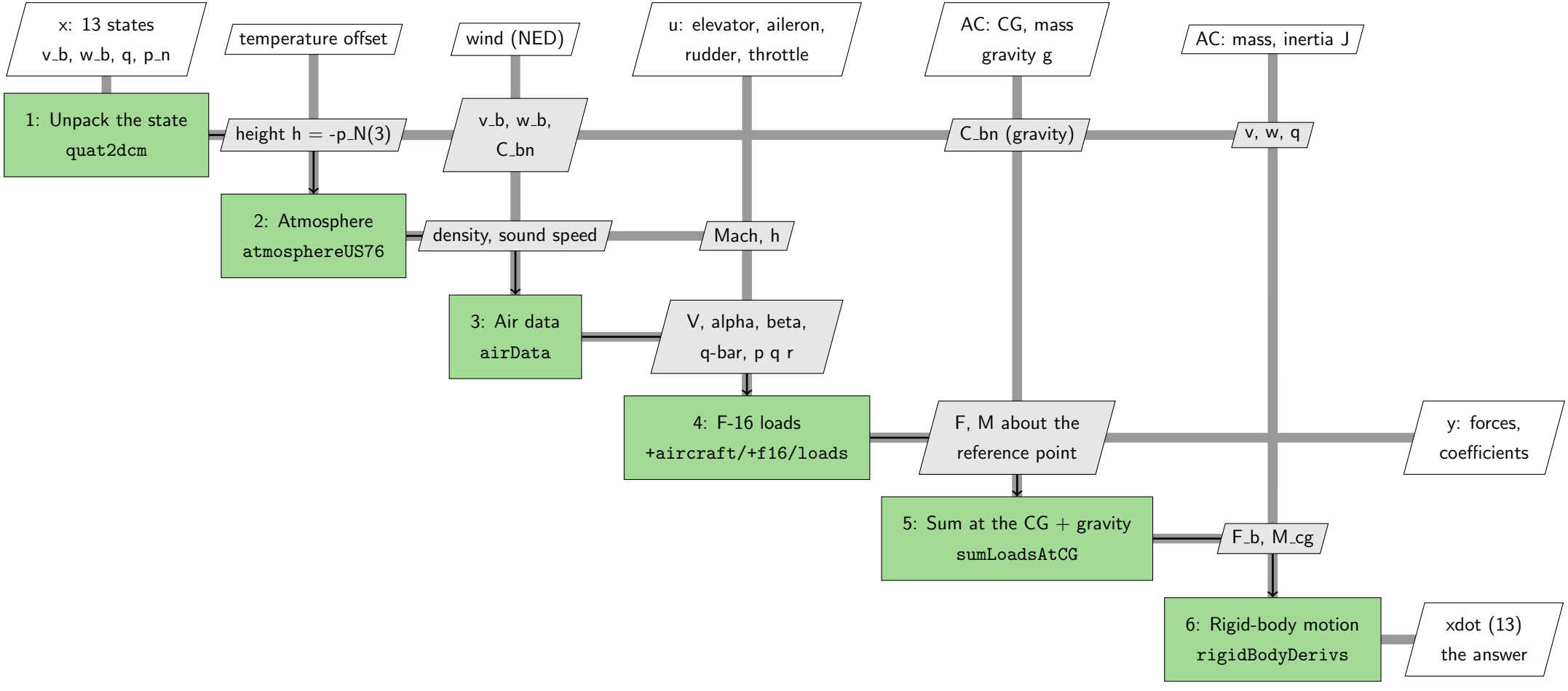
x: 13 states
v_b, w_b, q, p_n
temperature offset
wind (NED)
u: elevator, aileron,
rudder, throttle
AC: CG, mass
gravity g
AC: mass, inertia J
1: Unpack the state
quat2dcm
height h = -p_N(3)
v_b, w_b,
C_bn
C_bn (gravity)
v, w, q
2: Atmosphere
atmosphereUS76
density, sound speed
Mach, h
3: Air data
airData
V, alpha, beta,
q-bar, p q r
4: F-16 loads
+aircraft/+f16/loads
F, M about the
reference point
y: forces,
coefficients
5: Sum at the CG + gravity
sumLoadsAtCG
F_b, M_cg
6: Rigid-body motion
rigidBodyDerivs
xdot (13)
the answer